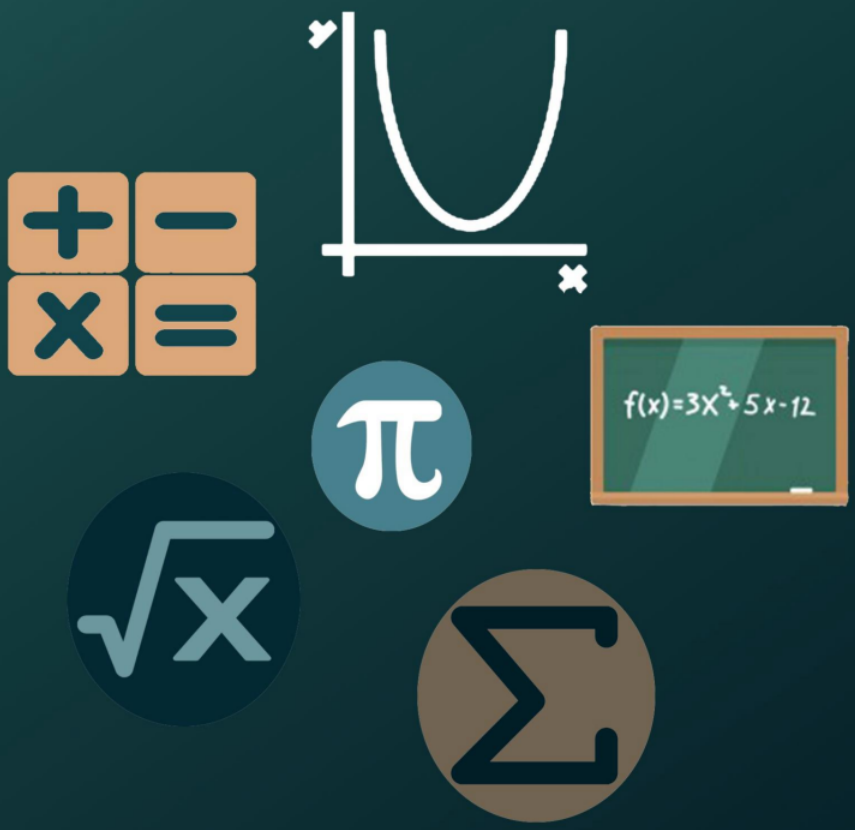
MATHEMATICS

Straight Line



SHEET

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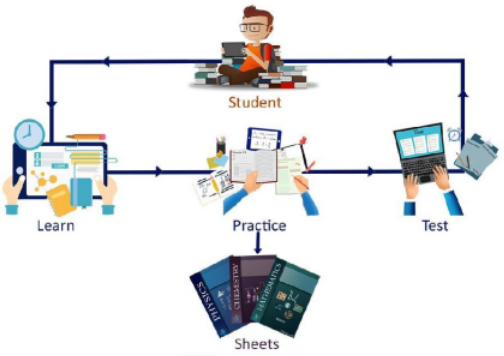
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Straight Line				
2	Standard Forms of Equations of Straight Lines - Part 1	1 - 5	1, 33	1	
3	Standard Forms of Equations of Straight Lines - Part 2	6 - 9	2, 34 - 36	2	23
4	Standard Forms of Equations of Straight Lines - Part 3	10	3,4, 22, 26,27, 37,38	3	1,17,24,25
5	Point and Line	11 - 12	5	4	2,18,19
6	Length of the Perpendicular from a Point on a Line	13	6,7, 28,29	5	26,27
7	The Angle between two Straight Lines	14 - 15	8,9,39		20
8	Equation of Lines - Parallel & Perpendicular to a given line	16 - 22	10		8-10,28
9	Equation of Straight Lines Equally Inclined to a given line	23 - 24	11	6	3
10	Family of Straight Lines		23 - 25	7-11	29
11	Concurrency of Straight Lines	25			14
12	Angle Bisectors	26	12,40	12	
13	Optics	27	13, 30, 41,42	13	21
14	Translation of the Axes	28 - 29			4,5
15	Pair of Straight Lines			14,15	6,7,30
16	General Equation of Second Degree representing a Pair of Straight Lines	30 - 31	14, 31	16-20	22
17	Homogenisation		32, 43 - 44		
18	After complete chapter		15 - 21	21-25	11-13,15, 16,31

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Standard Forms of Equations of Straight Lines - Part 1

Q1. If the line $y = mx + c$ passes through the points (2, 4) and (3, –5), then-

(A) $m = -9, c = -22$ (B) $m = 9, c = 22$

(C) $m = -9, c = 22$ (D) $m = 9, c = -22$

Q2. If the point (3, –4) divides the portion of line between the x and y axis in the ratio 2 : 3 then the equation of the line will be-

(A) $2x + y = 10$ (B) $2x - y = 10$

(C) $x + 2y = 10$ (D) $x - 2y = 10$

Q3. The equation to a line passing through the point (2, –3) and sum of whose intercept on the axes is equal to –2 is-

(A) $x + y + 2 = 0$ or $3x + 3y = 7$

(B) $x + y + 1 = 0$ or $3x - 2y = 12$

(C) $x + y + 3 = 0$ or $3x - 3y = 5$

(D) $x - y + 2 = 0$ or $3x + 2y = 12$

Q4. Equation to the straight line cutting off an intercept 2 from negative direction of the axis of y and inclined at 30° to the positive direction to the axis of x is :-

(A) $y + x - \sqrt{3} = 0$

(B) $y - x + 2 = 0$

(C) $y - \sqrt{3}x - 2 = 0$

(D) $\sqrt{3}y - x + 2\sqrt{3} = 0$

Q5. The angle between the lines $x - \sqrt{3}y + 5$ and y axis is-

(A) 90° (B) 60° (C) 30° (D) 45°

Standard Forms of Equations of Straight Lines - Part 2

Q6. The equation of a line passing through the point (–3, 2) and parallel to x-axis is-

(A) $x - 3 = 0$ (B) $x + 3 = 0$

(C) $y - 2 = 0$ (D) $y + 2 = 0$

Q7. The angle made by the line joining the points (1, 0) and $(-2, \sqrt{3})$ with x axis is-

(A) 120° (B) 60°

(C) 150° (D) 135°

Q8. The equation of the straight line which passess through the point (1, –2) and cuts off intercepts of equal length from axes is:-

(A) $x + y = 1$ (B) $x - y = 1$

(C) $x + y + 1 = 0$ (D) $x - y - 2 = 0$

Q9. The equation of the line inclined at an angle of 60° with x axis and cutting y axis at the point (0, –2) is-

(A) $\sqrt{3}y = x - 2\sqrt{3}$ (B) $y = \sqrt{3}x - 2$

(C) $\sqrt{3}y = x + 2\sqrt{3}$ (D) $y = \sqrt{3}x + 2$

Standard Forms of Equations of Straight Lines - Part 3

Q10. If line $3x + 3y + 7 = 0$ reduces to the form $x\cos\alpha + y\sin\alpha = p$ then the value of p is -

(A) $3\sqrt{7}/2$ (B) $7/2\sqrt{3}$

(C) $7/3\sqrt{2}$ (D) None

Point and Line

Q11. If the point (1,2) and (3,4) are to be on the opposite side of the line $3x - 5y + a = 0$, then-

(A) $7 < a < 11$ (B) $b = 7$

(C) $a = 11$ (D) $a < 7$ or $a > 11$

Q12. The position of the point (8, –9) with respect to the lines $2x + 3y – 4 = 0$ and $6x + 9y + 8 = 0$ is-

(A) point lies on same side of both the lines

(B) point lies on one of the lines

(C) point lies between the lines

(D) None of these

Length of the Perpendicular from a Point on a Line

Q13. The nearest point on the line $3x – 4y = 25$ from the origin is -

(A) (–4, 5) (B) (3, –4)

(C) (3, 4) (D) (3, 5)

The Angle between two Straight Lines

Q14. Number of integral values of m for which y-coordinate of point of intersection of lines $y = mx + 1$ & $x + y = 2$ is also an integer, is-

(A) 0 (B) 1 (C) 2 (D) 3

Q15. The angle between the lines $2x – y + 5 = 0$ and $3x + y + 4 = 0$ is-

(A) 30° (B) 90° (C) 45° (D) 60°

Equation of Lines - Parallel & Perpendicular to a given line

Q16. The equation of a line perpendicular to the line $5x – 2y + 7 = 0$ and passing through the point of intersection of lines $y = x + 7$ and $x + 2y + 1 = 0$, is -

(A) $2x + 5y = 0$ (B) $2x + 5y = 20$

(C) $2x + 5y = 10$ (D) None of these

Q17. If P is a point (x, y) on the line $y = –3x$ such that P and the point (3, 4) are on the opposite sides of the line $3x – 4y – 8 = 0$, then :-

(A) $x > \frac{8}{15}, y < -\frac{8}{5}$ (B) $x > \frac{8}{5}, y < -\frac{8}{15}$

(C) $x = \frac{8}{15}, y = -\frac{8}{5}$ (D) None of these

Q18. If the lines $mx + 2y + 1 = 0$ and $2x + 3y + 5 = 0$ are perpendicular then the value of m is-

(A) 3 (B) –3

(C) –1/3 (D) 1/3

Q19. A line is perpendicular to $3x + y = 3$ and passes through a point (2, 2). Its y intercept is-

(A) 2/3 (B) 1/3

(C) 1 (D) 4/3

Q20. The distance between the parallel lines $y = 2x + 4$ and $6x = 3y + 5$ is-

(A) $17/\sqrt{3}$ (B) 1

(C) $3/\sqrt{5}$ (D) $17\sqrt{5}/15$

Q21. The equation of a line passing through (a, b) and parallel to the line $x/a + y/b = 1$ is-

(A) $x/a + y/b = 0$ (B) $x/a + y/b = 2$

(C) $x/a + y/b = 3$ (D) $x/a + y/b + 2 = 0$

Q22. The equation of a line parallel to $x + 2y = 1$ and passing through the point of intersection of the lines $x – y = 4$ and $3x + y = 7$ is-

(1) $x + 2y = 5$ (2) $4x + 8y – 1 = 0$

(3) $4x + 8y + 1 = 0$ (4) None of these

Equation of Straight Lines Equally Inclined to a given line

Q23. The equation of the lines which passes through the point (3, –2) and are inclined at 60° to the line $\sqrt{3}x + y = 1$.

(A) $y + 2 = 0,\ \sqrt{3}x – y – 2 – 3\sqrt{3} = 0$

(B) $\sqrt{3}x – y – 2 – 3\sqrt{3} = 0$

(C) $x – 2 = 0,\ \sqrt{3}x – y + 2 + 3\sqrt{3} = 0$

(D) None of these

Q24. The equation of two equal sides of an isosceles triangle are $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side is passes through the point $(1, -10)$. The equation of the third side is :-

(A) $x - 3y - 31 = 0$ but not $3x + y + 7 = 0$

(B) neither $3x + y + 7 = 0$ nor $x - 3y - 31 = 0$

(C) $3x + y + 7 = 0$ or $x - 3y - 31 = 0$

(D) $3x + y + 7 = 0$ but not $x - 3y - 31 = 0$

Concurrency of Straight Lines

Q25. If the lines $y - x = 5$, $3x + 4y = 1$ and $y = mx + 3$ are concurrent then the value of m is :

(A) $\frac{19}{5}$ (B) 1 (C) $\frac{5}{19}$ (D) $\frac{1}{2}$

Angle Bisectors

Q26. The equation of the bisector of the angle between the lines $3x - 4y + 7 = 0$ and $12x - 5y - 8 = 0$ is -

(A) $99x - 77y + 51 = 0, 21x + 27y - 131 = 0$

(B) $99x - 77y + 51 = 0, 21x + 27y + 131 = 0$

(C) $99x - 77y + 131 = 0, 21x + 27y - 51 = 0$

(D) None of these

Optics

Q27. A ray of light coming from the point $(1, 2)$ is reflected at a point A on the x-axis and then passes through the point $(5, 3)$. The coordinates of the point A are-

(A) $(13/5, 0)$ (B) $(5/13, 0)$

(C) $(-7, 0)$ (D) None of these

Translation of the Axes

Q28. On shifting the origin to the point $(2, -5)$ and keeping the axis parallel, the new coordinates of the point $(-5, 3)$ will be :-

(A) $(7, -8)$

(B) $(-3, -2)$

(C) $(-7, 8)$

(D) none of these

Q29. If the axes are rotated through an angle of $30°$ in the anti-clockwise direction, the coordinates of points $(4, -2\sqrt{3})$ with respect to new axes are :-

(A) $(2, \sqrt{3})$ (B) $(\sqrt{3}, -5)$

(C) $(2, 3)$ (D) $(\sqrt{3}, 2)$

General Equation of Second Degree representing a Pair of Straight Lines

Q30. If the equation $12x^2 + 7xy - py^2 - 18x + qy + 6 = 0$ represents a pair of perpendicular straight lines, then-

(A) $p = 12, q = 1$

(B) $p = 1, q = -12$

(C) $p = 1, q = 12$

(D) $p = -1, q = 12$

Q31. If the equation $x^2 + y^2 + 2gx + 2fy + 1 = 0$ represent a pair of lines, then-

(A) $f^2 + g^2 = 1$

(B) $g^2 - f^2 = 1$

(C) $f^2 - g^2 = 1$

(D) $f^2 + g^2 = 1/2$

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Equation of lines passing through P(4, –3) and sum of whose intercepts on coordinates axes is 5, are :-

(A) $3x - 2y = 6$ (B) $x - 2y = 10$

(C) $3x + 2y = 6$ (D) $x + 2y = 10$

Q2. A line passes through points P(1, 4) & Q(–1, 3) then

(A) slope of the line is $\frac{1}{2}$

(B) x intercept of the line is –7

(C) y intercept of the line is 7

(D) y intercept is $\frac{7}{2}$

Q3. Coordinates of a point which is at +3 distance from points (1, –3) of line $2x + 3y + 7 = 0$ is :-

(A) $\left(1+\frac{9}{\sqrt{13}}, 3-\frac{6}{\sqrt{13}}\right)$

(B) $\left(1-\frac{9}{\sqrt{13}}, -3+\frac{6}{\sqrt{13}}\right)$

(C) $\left(1+\frac{9}{\sqrt{13}}, -3+\frac{6}{\sqrt{13}}\right)$

(D) $\left(1-\frac{9}{\sqrt{13}}, 3-\frac{6}{\sqrt{13}}\right)$

Q4. Normal form of line $x + \sqrt{3}y - 4 = 0$ is :-

(A) $x\cos\pi/3 - y\sin\pi/3 = 2$

(B) $x\cos\pi/6 - y\sin\pi/6 = 2$

(C) $x\cos\pi/3 + y\sin\pi/3 = 2$

(D) $x\cos\pi/6 + y\sin\pi/6 = 2$

Q5. Integral values of b for which origin and P(1, 1) lie on the same side of line $a^2x + aby + 1 = 0$ for all $a \in R-\{0\}$ are :-

(A) –2 (B) –1 (C) 0 (D) 1

Q6. Length of perpendicular from the origin to the line $3x + 4y = 10$ is :-

(A) 5 (B) 1 (C) 2 (D) 4

Q7. Co-ordinates of point on x-axis whose perpendicular distance from the straight line $\frac{x}{a}+\frac{y}{b}=1$ is 'a' :-

(A) $\left(\frac{a}{b}\left(b+\sqrt{a^2+b^2}\right), 0\right)$

(B) $\left(\frac{a}{b}\left(b-\sqrt{a^2-b^2}\right), 0\right)$

(C) $\left(\frac{a}{b}\left(b+\sqrt{a^2-b^2}\right), 0\right)$

(D) $\left(\frac{a}{b}\left(b-\sqrt{a^2+b^2}\right), 0\right)$

Q8. If $3x + 4y - 5 = 0$ and $4x + ky - 8 = 0$ are two parallel lines then k is :-

(A) 3/16 (B) 4/3

(C) 16/3 (D) –4

Q9. If $7x + 3y + 9 = 0$ and $y = kx + 7$ are two perpendicular lines then k is :-

(A) 3/7 (B) –7/3

(C) 3 (D) 7

Q10. A line which is perpendicular bisector at AB where A(4, 3) & B(2, –1) passes from the points :-

(A) (1, 2) (B) $\left(\frac{7}{2}, -4\right)$

(C) (–1, 3) (D) (9, –2)

Q11. One vertex of an equilateral triangle is (2, 3) and the equation of the line opposite to the vertex is x + y=2, then equation of remaining side is :-

(A) $y-3=(2+\sqrt{3})(x-2)$

(B) $y-3=(\sqrt{3}\pm1)(x-2)$

(C) $y-3=(2-\sqrt{3})(x-2)$

(D) None of these

Q12. Two lines are $3x-4y-5=0$ & $5x+12y-26=0$ then :-

(A) acute angle bisector is $64x+8y-195=0$

(B) acute angle bisector is $14x-112y+65=0$

(C) obtuse angle bisector contains the orgin

(D) acute angle bisector contains the origin

Q13. A line passing through A(3, 10) is reflected by line mirror 2x + y =6 at point P. Now reflected ray passes through point B(5, 6), then :-

(A) equation of incident rays is $4x-3y+18=0$

(B) equation of reflected ray is $y=6$

(C) equation of normal at P is $x-2y+12=0$

(D) area of ΔABP is 10 square units

Q14. If the angle between the pair of lines $2x^2+\lambda xy+3y^2+8x+14y+8=0$ is $\frac{\pi}{4}$ then value of λ is :

(A) 5 (B) 7

(C) –5 (D) –7

Q15. If the vertices P, Q, R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s)?

(A) centriod (B) incentre

(C) circumcentre (D) orthocentre

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 16 to 18)

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that AE = AF. Let P be a point inside the square ABCD.

Q16. The maximum possible area of quadrilateral CDFE is -

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

Q17. The value of $(PA)^2-(PB)^2+(PC)^2-(PD)^2$ is equal to -

(A) 3 (B) 2 (C) 1 (D) 0

Q18. Let a line passing through point A divides the square ABCD into two parts so that area of one portion is double the other, then the length of portion of line inside the square is

(A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{11}}{3}$ (D) $\frac{2}{\sqrt{3}}$

Paragraph # 2 (Ques. 19 to 21)

Let $M\left(2,\frac{13}{8}\right)$ is the circumcentre of ΔPQR whose sides PQ and PR are represented by the straight lines $4x-3y=0$ and $4x+y=16$ respectively.

Q19. The orthocentre of ΔPQR is -

(A) $\left(\frac{7}{3},\frac{4}{3}\right)$ (B) $\left(\frac{4}{3},\frac{7}{3}\right)$

(C) $\left(3,\frac{3}{4}\right)$ (D) $\left(\frac{3}{4},3\right)$

Q20. If A, B and C are the midpoint of the sides PQ, QR and PR of ΔPQR respectively, then the area of ΔABC equals -

(A) 1 (B) 2

(C) 3 (D) 4

Q21. If PB be the median of the ΔPQR, then the equation of the straight line passing through N(–2, 3) and perpendicular to PB is -

(A) $4x + y + 5 = 0$ (B) $x - 4y + 14 = 0$

(C) $4x - y + 11 = 0$ (D) $x + 4y - 10 = 0$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q22.

Column-I	**Column-II**
(A) Equation of Straight Line Which cuts-off an intercept 2 from the negative direction of the axis of y and inclined at 30° to OX.	(P) $x + y = 7$
	(Q) $10x + 7y = 11$
(B) Equation of Straight Line Which cuts-off equal interecepts on the co-ordinate axes and passes through (3, –1)	(R) $x + 4y = 8$
(C) Equation of Straight Line Passing to the points (–1, 3) and (6, –7)	(S) $3x - y = 2$
(D) Equation of Straight Line Through the point (4, 1) and making with the axes in the first quadrant a triangle whose area is 8.	(T) $x - \sqrt{3}y = 2\sqrt{3}$

Q23. The equation of the line through the intersection of the line $2x - 3y = 0$ and $4x - 5y = 2$ and

Column I	**Column II**
(A)Through the point (2, 1)	(P) $2x - y = 4$
(B) Perpendicular to line $x + 2y + 1 = 0$	(Q) $x + y - 5 = 0$, $x - y - 1 = 0$
(C) Parallel to line $3x - 4y + 5 = 0$	(R) $x - y - 1 = 0$
(D) Equally inclined to Axes	(S) $3x - 4y - 1 = 0$

Q24. Consider the line $Ax + By + C = 0$.

Match the nature of intercept of the line given in **column-I** with their corresponding conditions in **column-II**. The mapping is one to one only.

Column-I	**Column-II**
(A) x intercept is finite and y intercept is infinite	(P) $A = 0,\ B, C \neq 0$
(B) x intercept is infinite and y intercept is finite	(Q) $C = 0,\ A, B \neq 0$
(C) both x and y intercepts are zero	(R) $A, B = 0$ and $C \neq 0$
(D) both x and y intercepts are infinite	(S) $B = 0,\ A, C \neq 0$

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q25. The value of λ, if the line $(2x + 3y + 4) + \lambda(6x - y + 12) = 0$.

Column I	**Column II**
(P) is parallel to y-axis	(i) $\lambda = -\frac{11}{23}$
(Q) is perpendicular to $7x + y - 4 = 0$	(ii) $\lambda = -\frac{1}{3}$
(R) passes through point (2, 1)	(iii) $\lambda = -\frac{17}{41}$
(S) is parallel to x-axis	(iv) $\lambda = 3$

Then correct matching is :

(A) (P)-iv (Q)-ii (R)-iii (S)-i
(B) (P)-iv (Q)-iii (R)-i (S)-ii
(C) (P)-iv (Q)-i (R)-iii (S)-ii
(D) (P)-ii (Q)-i (R)-iv (S)-iii

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q26. Equation of a line which passes through point A (2, 3) and makes an angle of 45° with x axis. If this line meet the line $x + y + 1 = 0$ at point P then distance AP is a $\sqrt{2}$ units. Find the value of 'a'.

Q27. $7x + 24y = 175$ can be written as $x\cos\alpha + y\sin\alpha = P$ in normal form where α is the angle that normal from makes with x-axis and p is perpendicular distance of line from origin, then value of $\frac{P^2}{24\cot\alpha}$ is ?

Q28. Length of the perpendicular from the point (3, 4) on the line $3x + 4y + 10 = 0$ is ?

Q29. If perpendicular distance of point A(–3, –4) from straight line $12(x + 6) = 5(y - 2)$ can be written as $\frac{p}{q}$ where p & q are co-prime then $p - 5q$ is equal to **?**

Q30. The ordinate of a point P on the line $6x + y = 9$, which is closest to the point (–3, 1) can be expressed in the form a/b. Where $a, b \in N$ and are in lowest form, then find the value $(a + b)$.

Q31. If $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ representes two lines then λ is ?

Q32. Value of k for which pair of lines joining origin to points of intersection of curve $x^2 + y^2 + x - 2y - k = 0$ and line $x + y = 2$ are perpendicular.

SUBJECTIVE TYPE

Q33. Find the equation to the straight line which passes through the points (3,4) and have intercepts on the axes:
(i) equal in magnitude but opposite in sign
(ii) such that their sum is 14

Q34. Find the equation of the straight line bisecting the segment joining the points (5,3) and (4,4) and making an angle of 45° with the positive direction of x-axis.

Q35. Equation of a line passing through (3, –4) and (4, 3) is

Q36. The vertices of a triangle are A(4,–2), B(2,3) and C(5,–4). Find the equation of the median through C.

Q37. The length of perpendicular from the origin to a line is 9 and the line makes an angle of 120° with the positive direction of y-axis. Find the equation of the line.

Q38. Reduce $x + \sqrt{3}y + 4 = 0$ to the :
(i) Slope intercept form and find its slope and y-intercept
(ii) Intercept form and find its intercepts on the axes
(iii) Normal form and find the values of p and α.

Q39. Angle between $y = x + 5$ and $y = \sqrt{3}x - 4$ is ?

Q40. Given vertices A(1, 1), B(4, –2) & C(5, 5) of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

Q41. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.

Q42. Let P be the point (3, 2). Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.

Q43. Find angle and equation of bisectors of angle between the lines represented by the equation $2x^2 - 7xy + 3y^2 = 0$.

Q44. Find the combined equation of lines joining origin and points of intersection of circle $x^2 + y^2 = 4$ and $2x + 3y - 4 = 0$.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The equation of straight line passing through (–a, 0) and making the triangle with axes of area 'T' is
(A) $2Tx + a^2y - 2aT = 0$
(B) $2Tx - a^2y + 2aT = 0$
(C) $2T\,x - a^2y - 2aT = 0$
(D) None of these

Q2. The vertices B and C of a triangle ABC lie on the lines $3y = 4x$ and $y = 0$ respectively and the side BC passes through the point $\left(\frac{2}{3}, \frac{2}{3}\right)$. If ABOC is rhombus, O being the origin then equation of line BC is :-
(A) $y = -5x + 4$ (B) $2x + y = 2$
(C) $y = 4x - 2$ (D) None of these

Q3. $\frac{x}{a} + \frac{y}{b} = 1$ is a variable line such that $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{16}$. The locus of the foot of perpendicular from origin to the line is :-

(A) $x^2 + y^2 - ax - by = 0$
(B) $x^2 + y^2 + ax + by - a^2 - b^2 = 0$
(C) $x^2 + y^2 - 32 = 0$
(D) $x^2 + y^2 - 16 = 0$

Q4. The range of values of α in the interval $(0, \pi)$ such that points (3, 5) and $(\sin\alpha, \cos\alpha)$ lie on the same side of the line $x + y - 1 = 0$ is

(A) $\left(0, \frac{\pi}{4}\right)$ (B) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(C) $\left(0, \frac{\pi}{2}\right)$ (D) None

Q5. If p and q are the lengths of perpendiculars from the origin to the lines $x\cos\theta - y\sin\theta = K\cos2\theta$ and $x\sec\theta + y\,\text{cosec}\theta = K$, respectively, then:-

(A) $p^2 + q^2 = K^2$ (B) $4p^2 + q^2 = K^2$

(C) $p^2 + 4q^2 = K^2$ (D) $p^2 + q^2 = \frac{K^2}{2}$

Q6. A straight line through the point (2, 2) intersects the lines $x + \sqrt{3}y = 0$ and $x - \sqrt{3}y = 0$ at the points A and B. The equation to the line AB so that the triangle OAB is equilateral, is

(A) $x - 2 = 0$ (B) $y - 2 = 0$
(C) $x + y - 4 = 0$ (D) None of these

Q7. Two consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, then the equation to the other diagonal is :-

(A) $3x + 7y = 0$ (B) $11x - 7y = 0$
(C) $x - y = 0$ (D) $x + y = 0$

Q8. Given the family of lines, $a(3x + 4y + 6) + b(x + y + 2) = 0$. The line of the family situated at the greatest distance from the point P (2,3) has equation -

(A) $4x + 3y + 8 = 0$
(B) $5x + 3y + 10 = 0$
(C) $15x + 8y + 30 = 0$
(D) none

Q9. Find the fix point through which the line $x(a + 2b) + y(a + 3b) = a + b$ always passes for all values of a and b-

(A) (2, 1) (B) (1, 2)
(C) (2, –1) (D) (1. –2)

Q10. Find the fix point through which the line $(2\cos\theta + 3\sin\theta)x + (3\cos\theta - 5\sin\theta)y - (5\cos\theta - 2\sin\theta) = 0$ passes for all values of θ.

(A) (0, 0) (B) (1, 1)
(C) (2, 1) (D) None of these

Q11. The straight lines $ax + by + c = 0$ where $3a + 2b + 4c = 0$ are concurrent at the point :-

(A) (1/2, 3/4) (B) (3/4, 1/2)
(C) (–3/4, –1/2) (D) (–3/4, 1/2)

Q12. Let P(–1, 0), Q(0, 0) and R(3, $3\sqrt{3}$) be three points. Then the equation of the bisector of the angle $\angle$ PQR is -

(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3}y = 0$

(C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

Q13. The reflection of A(3, 4) in the line $x - 2y + 10 = 0$ is the point B. The reflection of B in line $x - y + 1 = 0$ is the point C, then the circumcenter of the triangle ABC is-

(A) (5, 6) (B) (8, 9)
(C) (7, 8) (D) None of these

Q14. If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ be the square of the other, then $\frac{a+b}{h} + \frac{8h^2}{ab}$ is -

(A) 3 (B) 4 (C) 5 (D) 6

Q15. If the angle between lines represented by equation $y^2 + kxy - x^2\tan^2A = 0$ is 2A then k =

(A) 0 (B) 2 (C) 4 (D) –2

Q16. If the angle between the lines represented by

$6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$

is $\tan^{-1}(m)$ and $a^2 + b^2 - ab - a - b + 1 \le 0$, then $5a + 6b$ is equal to $(a, b \in R)$

(A) 1/m (B) m

(C) 1/2m (D) 2m

Q17. If the acute angle between the pairs of lines $3x^2 - 7xy + 4y^2 = 0$ and $6x^2 - 5xy + y^2 = 0$ are θ_1 and θ_2 respectively, then-

(A) $\theta_1 = \theta_2$ (B) $2\theta_1 = \theta_2$

(C) $\theta_1 = 2\theta_2$ (D) None of these

Q18. Equation of the pair of straight lines through origin and perpendicular to the pair of straight lines $5x^2 - 7xy - 3y^2 = 0$ is-

(A) $3x^2 - 7xy - 5y^2 = 0$

(B) $3x^2 + 7xy + 5y^2 = 0$

(C) $3x^2 - 7xy + 5y^2 = 0$

(D) $3x^2 + 7xy - 5y^2 = 0$

Q19. The points of intersection of lines represented by the curve $2x^2 + 3xy + y^2 - 7x - 5y + 6 = 0$ is-

(A) (1, –1) (B) (2, 1)

(C) (1, 1) (D) None of these

Q20. If slope of lines represented by $ax^2 + 2hxy + by^2 = 0$ are in 1 : 3 ratio then $h^2 : ab$ equals-

(A) 3/4 (B) 1/4

(C) 4/3 (D) 1

Q21. A variable straight line L with negative slope passes through the point (8, 2) and cuts the positive co-ordinate axes at points P and Q. The absolute minimum value of OP + OQ, where O is origin.

(A) 16 (B) 18

(C) 14 (D) 20

Q22. The graph of the function

$y = \cos x \cos(x + 2) - \cos^2(x + 1)$ is

(A) a straight line passing through $(0, -\sin^2 1)$ with slope 2

(B) a straight line passing through (0, 0)

(C) a parabola with vertex $(1, -\sin^2 1)$

(D) a straight line passing through the point $\left(\frac{\pi}{2}, -\sin^2 1\right)$ and parallel to x-axis

Q23. A line passing through the point (2, 2) and the axes enclose an area λ. The intercepts on the axes made by the line are given by the two roots of

(A) $x^2 - 2|\lambda|x + |\lambda| = 0$

(B) $x^2 + |\lambda|x + 2|\lambda| = 0$

(C) $x^2 - |\lambda|x + 2|\lambda| = 0$

(D) none of these

Q24. (1, 2) is vertex of a square whose one diagonal is along the x-axis. The equations of sides passing through the given vertex are-

(A) $2x - y = 0, x + 2y + 5 = 0$

(B) $x - 2y + 3 = 0, 2x + y - 4 = 0$

(C) $x - y + 1 = 0, x + y - 3 = 0$

(D) None of these

Q25. Consider the point $A \equiv (0,1)$, $B \equiv (2,0)$ and 'P' be a point on the line $4x + 3y + 9 = 0$. Coordinates of the point 'P' such that $|PA - PB|$ is maximum is-

(A) $\left(-\frac{12}{5}, \frac{17}{5}\right)$ (B) $\left(-\frac{84}{5}, \frac{13}{5}\right)$

(C) $\left(-\frac{6}{5}, \frac{17}{5}\right)$ (D) $\left(-\frac{24}{5}, \frac{17}{5}\right)$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x - \sqrt{3}y = 0$, then the co-ordinates of the third vertex are

(A) (0, a) (B) $\left(\frac{\sqrt{3}\ a}{2}, -\frac{a}{2}\right)$

(C) (0, –a) (D) $\left(-\frac{\sqrt{3}\ a}{2}, \frac{a}{2}\right)$

Q2. All points lying inside the triangle formed by the points (1, 3), (5, 0) and (–1, 2) satisfy-

(A) $3x + 2y \geq 0$

(B) $2x + y - 13 \geq 0$

(C) $2x - 3y - 12 \leq 0$

(D) $-2x + y \geq 0$

Q3. A (1, 3) and C (7, 5) are two opposite vertices of a square. The equation of a side through A is

(A) $x + 2y - 7 = 0$ (B) $x - 2y + 5 = 0$

(C) $2x + y - 5 = 0$ (D) $2x - y + 1 = 0$

Q4. If line $y - x + 2 = 0$ is shifted parallel to itself by a perpendicular distance of $3\sqrt{2}$ units, then the equation of the new line is may be -

(A) $y = x + 4$

(B) $y = x + 1$

(C) $y = x - (2 + 3\sqrt{2})$

(D) $y = x - 8$

Q5. Equation of pair of lines passing through (1, –1) and parallel to the lines $2x^2 + 5xy + 3y^2 = 0$ is -

(A) $2(x-1)^2 + 5(x-1)(y+1) + 3(y+1)^2 = 0$

(B) $3(x-1)^2 - 5(x-1)(y+1) + 2(y+1)^2 = 0$

(C) $2x^2 + 5xy + 3y^2 + x + y = 0$

(D) $3x^2 - 5xy + 2y^2 - 11x + 9y + 10 = 0$

Q6. The pairs of straight lines $ax^2+2hxy-ay^2=0$ and $hx^2 - 2axy - hy^2 = 0$ are such that-

(A) one pair bisects the angles between the other pair

(B) the lines of one pair are equally inclined to the lines of the other pair

(C) the lines of one pair are perpendicular to the lines of the other pair

(D) none of these

Q7. If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of lines whose slopes are m and m^2, then value of a is/are

(A) $a = -8$ (B) $a = 8$

(C) $a = 27$ (D) $a = -27$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Passage # 1 (Ques 8 to 10)

$L_1 : 3x + 4y + 8 = 0$

$L_2 : 2x + 7y - 1 = 0$

Q8. If L_1, L_2 represent the sides AB and AC of the isosceles triangle ABC with AB = AC = 2 then the co-ordinates of -

(A) B are $\left(\frac{28}{5}, -\frac{11}{5}\right)$

(B) B are $\left(\frac{28}{5}, \frac{1}{5}\right)$

(C) C are $\left(\frac{14+4\sqrt{53}}{\sqrt{53}}, \frac{-4+\sqrt{5}}{\sqrt{5}}\right)$

(D) C are $\left(\frac{14+4\sqrt{53}}{\sqrt{53}}, \frac{4-\sqrt{5}}{\sqrt{5}}\right)$

Q9. Equation of the line through B parallel to AC is -

(A) $2x + 7y + 21 = 0$

(B) $10x + 35y = 63$

(C) $10x + 35y + 21 = 0$

(D) $2x + 7y = 63$

Q10. If D is the mid-point of BC and E is the mid-point of CA then DE is equal to -

(A) 1/4 (B) 1/2 (C) 1 (D) 2

Passage # 2 (Ques 11 to 13)

Consider two points $A \equiv (1, 2)$ and $B \equiv (3, -1)$. Let M be a point on the straight line $L \equiv x + y = 0$.

Q11. If M be a point on the line $L = 0$ such that $AM + BM$ is minimum, then the reflection of M in the line $x = y$ is

(A) $(1, -1)$ (B) $(-1, 1)$

(C) $(2, -2)$ (D) $(-2, 2)$

Q12. If M be a point on the line $L = 0$ such that $|AM - BM|$ is maximum, then the distance of M from $N \equiv (1, 1)$ is -

(A) $5\sqrt{2}$ (B) 7

(C) $3\sqrt{5}$ (D) 10

Q13. If M be a point on the line $L = 0$ such that $|AM - BM|$ is minimum, then the area of ΔAMB equals

(A) $\frac{13}{4}$ (B) $\frac{13}{2}$

(C) $\frac{13}{6}$ (D) $\frac{13}{8}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. Consider the lines given by

$L_1 : x + 3y - 5 = 0$, $L_2 : 3x - ky - 1 = 0$, $L_3 : 5x + 2y - 12 = 0$

Now match the following :

Column-I	**Column-II**
(A) L_1, L_2, L_3 are concurrent, if	(P) $k = -9$
(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(Q) $k = -\frac{6}{5}$
(C) L_1, L_2, L_3 form a triangle, if	(R) $k = \frac{5}{6}$
(D) L_1, L_2, L_3 do not form a triangle, if	(S) $k = 5$

Q15.

Column-I	**Column-II**
(A) If the lines $ax + 2y + 1 = 0$; $bx + 3y + 1 = 0$ & $cx + 4y + 1 = 0$ are concurrent then $a + c - 2b$ is equal to.	(p) 1
(B) If line $2x - by + 1 = 0$ intersects the curve $2x^2 - by^2 + (2b - 1)xy - x - by = 0$ at points A & B and AB subtends a right angle at origin, then value of $b + b^2$ is equal to	(q) 0
(C) A line passes through point (3, 4) and the point of intersection of the lines $4x + 3y = 12$ and $3x + 4y = 12$ and length of intercepts on the co-ordinate axes are a and b, then ab is equal to	(r) 5
(D) A light ray emerging from the point source placed at P(2, 3) is reflected at a point 'Q' on the y-axis and then passes through the point R(5, 10). If co-ordinates of Q are (a, b), then $a + b$ is	(s) 4

Q16. Let ABC be a triangle such that the coordinates of A are (– 3, 1). Equation of the median through B is $2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Then match the entries of column-I with their corresponding correct entries of column-II.

Column-I	**Column-II**
(A) Equation of the line AB is	(P) $2x + y - 3 = 0$
(B) Equation of the line BC is	(Q) $2x - 3y + 9 = 0$
(C) Equation of CA is	(R) $4x + 7y + 5 = 0$
	(S) $18x - y - 49 = 0$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. If the equations of the diagonals of the parallelogram formed by the lines,
$2x - y + 7 = 0$; $2x - y - 5 = 0$;
$3x + 2y - 5 = 0$ and $3x + 2y + 4 = 0$
are $ax + by - 5 = 0$ and $px + qy + 1 = 0$.
Find the value of $(a + b + p + q)$.

Q18. If $P(\sin\theta, \cos\theta)$, $\theta \in [0, 2\pi)$ & $Q(1, 4)$ lie on the same side or on the line $\sqrt{3}x - y + 1 = 0$ then maximum value of $\sin\theta$ is b. Value of $4b^2$ is ?

Q19. If (t, t^2) falls inside the angle made by the lines $y = x/2$, $x > 0$ and $y = 3x$, $x > 0$; where $t \in (a, b)$ then find the value of $2a + b$.

Q20. Consider 3 lines :
L_1 : $5x - y + 4 = 0$
L_2 : $3x - y + 5 = 0$
L_3 : $x + y + 8 = 0$
If these lines enclose a triangle ABC and sum of the squares of the tangents of the interior angles can be expressed in the form p/q where p and q are relatively prime numbers, compute the value of $(p + q)$.

Q21. A ray of light is sent along the line $x - 2y - 3 = 0$. Upon reaching the line $3x - 2y - 5 = 0$, the ray is reflected from it. If the equation of the line containing the reflected ray is expressed in the form $ax - by - c = 0$. Given that a, b and c are positive integers. Find the value of $a + b + c$.

Q22. The equation
$9x^3 + 9x^2y - 45x^2 = 4y^3 + 4xy^2 - 20y^2$
represents 3 straight lines, two of which pass through the origin. Find the area of the triangle formed by these lines (in sq. units).

SUBJECTIVE TYPE

Q23. The line $3x + 2y = 24$ meets the y-axis at A & the x-axis at B. The perpendicular bisector of AB meets the line through $(0, -1)$ parallel to x-axis at C. Find the area of the triangle ABC.

Q24. Let O(0, 0), A(6, 0) and B(3, $\sqrt{3}$) be the vertices of ΔOAB. Let R be the region consisting of all those points P inside ΔOAB which satisfy $d(P, OA) \leq$ minimum $(d(P,OB), d(P,AB))$ where d(P,OA), d(P,OB) and d(P,AB) represent the distance of P from the sides OA,OB and AB respectively. If the area of region R is $9(a - \sqrt{b})$ where a and b are coprime, then find the value of $(a + b)$.

Q25. A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point (1, 5). Obtain the equation.

Q26. The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.

Q27. The equations of the perpendicular bisectors of the sides AB & AC of a triangle ABC are $x - y + 5 = 0$ & $x + 2y = 0$, respectively. If the point A is (1, –2) find the equation of the line BC.

Q28. Two sides of a rhombus ABCD are parallel to the lines $y = x + 2$ & $y = 7x + 3$. If the diagonals of the rhombus intersect at the point (1, 2) & the vertex A is on the y-axis, find the possible coordinates of A.

Q29. Consider the family of lines
$(x - y - 6) + \lambda(2x + y + 3) = 0$
and $(x + 2y - 4) + \lambda(3x - 2y - 4) = 0$.
If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.

Q30. The distance of a point (x_1, y_1) from each of two straight lines which passes through the origin of co-ordinates is δ ; find the combined equation of these straight lines.

Q31. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the ΔABC are G, H and S respectively. $L = 0$ denotes the equation of sides BC.

(a) If $L_1 : 2x - y = 0$ and $L_2 : x + y = 3$ and G(2, 3) then find the slope of the line $L = 0$.

(b) If $L_1 : 2x + y = 0$ and $L_2 : x - y + 2 = 0$ and H(2, 3) then find the y-intercept of $L = 0$.

(c) If $L_1 : x + y - 1 = 0$ and $L_2 : 2x - y + 4 = 0$ and S(2, 1) then find the x-intercept of the line $L = 0$.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
C	B	B	D	B	C	C	C	B	C
11	12	13	14	15	16	17	18	19	20
A	A	B	C	C	A	A	B	D	D
21	22	23	24	25	26	27	28	29	30
B	B	A	C	C	A	A	C	B	A
31									
A									

Ex - 1A

1. B,C **2.** A,B,D **3.** B **4.** C **5.** B,C,D
6. C **7.** A,D **8.** C **9.** A **10.** A,C,D
11. C, A **12.** B,D **13.** A, B, C, D **14.** B,D **15.** A,C,D
16. C **17.** D **18.** B **19.** C **20.** B
21. D
22. A → T, B → P, C → Q, D → R **23.** A → R, B → P, C → S, D → Q
24. A → S, B → P, C → Q, D → R
25. B **26.** 3 **27.** 7 **28.** 7 **29.** 1

30. 100 **31.** 2 **32.** 3 **33. (i)** $x - y + 1 = 0$, **(ii)** $4x + 3y = 24$ and $x + y = 7$

34. $x - y - 1 = 0$ **35.** $y = 7x - 25$ **36.** $9x + 4y - 29 = 0$ **37.** $x + y\sqrt{3} = 18$

38. **(i)** $y = \left(-\frac{1}{\sqrt{3}}\right)x + \left(-\frac{4}{\sqrt{3}}\right)$ whose slope (m) $= -\frac{1}{\sqrt{3}}$ and y- intercept (c) $= -\frac{4}{\sqrt{3}}$

(ii) $\frac{x}{-4} + \frac{y}{-4/\sqrt{3}} = 1$ where x-intercept (a) = –4 and y-intercept (b) $= -\frac{4}{\sqrt{3}}$

(iii) $\left(-\frac{1}{2}\right)x + \left(-\frac{\sqrt{3}}{2}\right)y = 2$ $\alpha = 240°$, $p = 2$

$\therefore$ Required normal form is $x \cos 240° + y \sin 240° = 2$

39. 15° or 165° **40.** $x - 5 = 0$ **41.** $2x - y + 3 = 0$, $2x + y - 7 = 0$, $x - 2y - 6 = 0$

42. 15 **43.** 45° ; $7x^2 - 2xy - 7y^2 = 0$ **44.** $5y^2 + 12xy = 0$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**B**	**D**	**C**	**C**	**A**	**C**	**A**	**C**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**C**	**B**	**D**	**A**	**D**	**A**	**A**	**C**	**C**
21	**22**	**23**	**24**	**25**					
B	**D**	**C**	**C**	**D**					

EX - 2A

1. A,B,C,D **2.** A,C **3.** A,D **4.** A,D **5.** A,C

6. A,B **7.** B,D **8.** A **9.** C **10.** C

11. B **12.** D **13.** A

14. A → S,B → P,Q, C → R, D → P,Q,S

15. (A)→(Q); (B)→(S); (C)→(P); (D)→(R)

16. A→R, B→S , C→Q

17. 40 **18.** 3 **19.** 4 **20.** 465 **21.** 62

22. 30 **23.** 91 sq.units **24.** 5 **25.** $83x - 35y + 92 = 0$

26. $c = -4$; B(2, 0); D(4, 4) **27.** $14x + 23y = 40$ **28.** (0, 0) or $\left(0, \frac{5}{2}\right)$

29. $x^2 + y^2 - 3x + 4y - 3 = 0$ **30.** $(y_1^2 - \delta^2)\, x^2 - 2x_1y_1xy + (x_1^2 - \delta^2)\, y^2 = 0$

31. (a) 5; (b) 2; (c) 3/2

-: Important Notes & Formulas :-